

KENDRIYA VIDYALAYA SANGATHAN BHUBANESWAR REGION

PRE- BOARD EXAMINATION (2024-25)

CLASS – X

SUBJECT: MATHEMATICS, STANDARD (CODE – 041)

TIME -3HOURS

M.M: 80

MARKING SCHEME

S.No.	SOLUTION	MARKS
SECTION A		
1	Option (d) none of these (answer- $3/2$ and $-1/3$)	1
2	(a) 2:1	1
3	(c) $8 - 3n$	1
4	(d) 140π	1
5	(b) -10	1
6	(b) 12 cm	1
7	(d) 35	1
8	(c) 5	1
9	(d) -25	1
10	(b) $2\sqrt{a^2 + b^2}$	1
11	b) $\sqrt{6} : \sqrt{\pi}$	1
12	(b) 1	1
13	a) $12/13$	1
14	(d) $\cos A$	1
15	(a) 0.77	1
16	(c) 6cm	1
17	(c) 120°	1
18.	(c) 0	1
19	(a)	1
20	(c)	1
SECTION B [2 X 5 =10]		
21.	Let the ratio be k:1 Then $3k-6/k+1 = -4$ $K = 2/7$ Or, k:1= 2:7 $y=2x(-8)+7 \times 10 /9= 54/9=6$ OR Let the ratio be k:1 and any point on Y axis is (0,y) Then $5k-1 = 0$ $k = 1/5$ Or, k:1= 1:5 $y=(-6)x(1)+5x(-4) /6= -26/6 = -13/3$ point of intersection = $\left(0, -\frac{13}{3}\right)$ —	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ OR $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

22.	Given AB = AC $(5-0)^2 + (-3-1)^2 = (x-0)^2 + (6-1)^2$ $25+16=x^2+25$ $x = \pm 4$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
23	$2(1)^2 + 2(1/\sqrt{2})^2 - 2(1/\sqrt{2})^2 + (1)^2 = 3$ OR $4 + 4 \tan^2 A$ $= 4 (1 + \tan^2 A)$ $= 4 (\sec^2 A) = 4 (5/2)^2 = 25$	1 for all correct values 1 1 $\frac{1}{2}$ $\frac{1}{2}$
24	LCM of 60 , 84 = 420 Reqd. No.= 420+2 =422	1 1
25.	i. $6/52=3/26$ ii. $8/52 = 2/13$	1 1
SECTION C [3 X 6 = 18]		
26	$(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2$ $\sin^2 A + \operatorname{cosec}^2 A + 2 \sin A . \operatorname{cosec} A + \cos^2 A + \sec^2 A + 2 \cos A . \sec A$ $(\sin^2 A + \cos^2 A) + \operatorname{cosec}^2 A + \sec^2 A + 2 + 2$ $1 + 1 + \cot^2 A + 1 + \tan^2 A + 2 + 2$ $= 7 + \cot^2 A + \tan^2 A = \text{RHS}$ OR $\sec A (1 - \sin A)(\sec A + \tan A)$ $= (\sec A - \sec A \times \sin A) (\sec A + \tan A)$ $= (\sec A - \tan A) (\sec A + \tan A)$ $= (\sec^2 A - \tan^2 A) = 1$	1 for expansion 1 for conversion 1 1 1 1
27.	figure LM CB implies AM/BM = AL/LC By BPT ----1 and LN CD implies AL/LC = AN/ND By BPT-----2 From 1 & 2 AM/BM = AN/ND Implies AM/AN = BM/DN By alternendo Or DE AC implies BD/AB = BE/BC1 By BPT DF AE implies BD/AB = BF/BE2 By BPT From 1 & 2 BE/BC = BF/BE implies $BE^2 = BF \cdot BC$	1 1 $\frac{1}{2}$ $\frac{1}{2}$ OR 1 1 1
28	Correct proof	3
29..	$r = 21, \theta = 60$ i.Length = $\theta \times 2\pi r/360 = 60 \times 2 \pi \times 21 / 360 = 22 \text{ cm}$ ii.Area = $\theta \times \pi r^2/360 = 60 \times \pi \times 21 \times 21 / 360 = 21 \times 11 = 231 \text{ sq. cm}$	$\frac{1}{2}$ for correct formula and 1 for correct answer each

30	Let the two parts are x , $27-x$ $\frac{1}{x} + \frac{1}{27-x} = \frac{3}{20}$ $140 = 3x(27-x) = 81x - 3x^2$ $3x^2 - 81x + 540 = 0$ $x^2 - 27x + 180 = 0$ $X=15$ $x=12$ Or Let the unit place digit is x and the other digit is $18/x$ ATQ $10x + \frac{18}{x} - 63 = 10\left(\frac{18}{x}\right) + x$ $10x^2 + 18 - 63x = 180 + x^2$ $9x^2 - 63x - 162 = 0$ $x^2 - 7x - 18 = 0$ $X=9$, -2 number = 92	$\frac{1}{2}$ $\frac{1}{2}$ 1 1 $\frac{1}{2}$ $\frac{1}{2}$ 1 1
² 31	Let one zero be α , Given $a = a^2 + 9$, $b = 13$, $c = 6a$ Product of zeroes = 1 implies $c/a = 1$, $6a = a^2 + 9$ $a^2 - 6a + 9 = 0$ $a = 3$ Or, Sum of Zeroes of $2x^2 + 7x + 6$ is $-7/2$ Product = 3 $3\alpha + 3\beta = -21/2$ product = $3\alpha \cdot 3\beta = 9 \times 3 = 27$ Polynomial $x^2 - 21x/2 + 27 = 2x^2 - 21x + 54$	$\frac{1}{2}$ $\frac{1}{2}$ 1 1 1 1 1
SECTION D [5 X 4 =20]		
32	Median = 28.5 Median Class = 20-30 From the table, $45 + x + y = 60$ $x + y = 60 - 45$ $x + y = 15$ $Median = l + (N/2 - cf) h / f$ $28.5 - 20 = (30 - 5 - x)/2$ $x = 8$ $x + y = 15$ $8 + y = 15$ $y = 15 - 8$, $y = 7$ $\therefore x = 8$, $y = 7$	Correct table 2 marks. 1 1 1
33	Given $-\frac{1}{2}$; Figure $-\frac{1}{2}$; To prove $-\frac{1}{2}$ Correct Proof – 1.5 marks Fig $\frac{1}{2}$ Correct Proof with reason – 1.5 marks OR Naming of points and joining OB , OC , OD QD = QB = 8 , DR = CR = 6 , PB = PC = x say PQ = 8 + x , QR = 14 , PR = 6 + x $s = 14 + x$ Area PQR = Area OPQ + Area OQR + area OPR Solving for $x = 7$ PQ = 15 , PR = 13	1.5 1.5 0.5 1.5 1/2 1/2 1 2 + 1 for formula
34	Let speed of A is x and speed of B is y km/hr, fig $5x - 5y = 100$ Or, $x - y = 20$ $x + y = 100$ solving $x = 60$ km/hr , $y = 40$ km/hr Or, Table Drawing each graph with shading of graph Finding point of intersection (1,4) , $x = 1$, $y = 4$ Finding area with Y axis = $\frac{1}{2} \times 4 \times 1 = 2$ sq. units	1 1 1 2 1 2 1 1

35	Correct fig $\tan 30^\circ = CD/AD$ Or, $cd/ad = 1/\sqrt{3}$ Or, $AD = \sqrt{3} CD$ $\tan 60^\circ = CD/BD$ Or, $CD/BD = \sqrt{3}$ Or, $BD = CD/\sqrt{3} = \sqrt{3} CD/3$ Given $AD-BD = 20$ OR, $2\sqrt{3} CD/3 = 20$ $CD = 10\sqrt{3}$ HEIGHT $CD = 10\sqrt{3}$ Distance $AD = 30$	1 1 1 1 1
SECTION E [4 X 3 =12]		
36.	(1) $\sqrt{3}$ m (2) Amount of air inside tent = Volume of tent = πr^2 height of cylinder + πr^2 height of cone/3 = $4.84\pi \times (2.1 + 1/\sqrt{3})$ cubic m (3) SA = 48.4 sq.m Cost = 24200 rupees OR SA = 48.4 sq.m weight = 96.8 kg	1 1 1+1 1+1
37	i.16 ii.5(correct formula and correct substitution 1 mark & Right answer 1 mark) Or, No. of pipes = $S_n - (a + a_n) = 175$ (iii) 20	1 1+1=2 1+1=2 1
38	(i) $40/8 = x/30 = 150m$ (ii) Difference of shadows = 1.2 m Shadow of house = $30 - 1.2 = 28.8$ Height of house = $28.8 \times 5 = 144m$ Difference of house = $150 - 144 = 6m$ Or, Shadow of larger house = $2 \times 3 = 6m$ (ratio of height = ratio of shadow) (iii) $14/5.6 = 70/x$ $X = 28cm$ difference = 28 cm $- 5.6cm = 22.4cm$	1 1+1 1+1 1